

Lab 1 - DS

- 0) $\boxed{;}$ $\rightarrow$ executes and prints
 $\boxed{:}$ $\rightarrow$ executes without printing

$\boxed{\text{restart;}}$ $\rightarrow$ clear memory

evalf , eval , expand , solve , plot , plot3d , implicitplot .

- 1) Floating point evaluation $\rightarrow$ use decimals (evalf) instead of $\left\{ \begin{array}{l} \text{symbols} \\ \text{fractions} \dots \end{array} \right.$

$\boxed{\text{evalf}}$ $\rightarrow$ gives decimal approximation $\rightarrow$ decimal output

$\boxed{\text{eval}}$ $\rightarrow$ exact value $\rightarrow$ symbols, radicals, fractions, Pi ...

$$\text{eval}(\sin(0.1)) = \text{evalf}(\sin(0.1)) = 0.09983341665$$

$$\text{eval}(\sin(\text{Pi}/4)) \neq \text{evalf}(\sin(\text{Pi}/4))$$

$$\downarrow$$
$$\frac{\sqrt{2}}{2}$$

$$\downarrow$$
$$0.7071067812$$

- 2) Assign , expand , unassign

$\boxed{:=}$ $\rightarrow$ assign

$\boxed{\text{expand}}$ $\rightarrow$ expands products/powers/parenthesis...

$\boxed{a := 'a'}$ $\rightarrow$ unassign

- 3) Factorize

$\boxed{\text{factor}}$ $\rightarrow$ factors polynomials

(big expression $\Rightarrow$ expression with parenthesis).

- 4) Add rational expressions

$\boxed{\text{factor}}$ $\rightarrow$ $\nexists$ simplifies rational expressions by factoring denominators.

5) Simplify a trigon. expression.

simplify (...., trig)

6) Evaluate an expression.

subs (x=..., expr) → "substitutes" → exact value (symbols)

≠ eval (expr, x=1) → "evaluates" → "substitutes" + "simplifies"

ex: subs (x=1, exp(x) + ln(x)); → $e + \ln(1)$.

eval (exp(x) + ln(x), x=1); → $e + \underbrace{\ln(1)}_0 = e$.

7) Solve equations / systems.

solve → exact solution (symbolic)

≠ fsolve → when "solve" gets "root of..." → numeric sol.

ex: solve (x^2 - 4*x + 3 = 0, x);

7d) fsolve (x - cos(x) = 0, x = -1..1);

7e) fsolve (x^5 - 3*x^3 - 1 = 0, x = -2..2);

7f) SYSTEM:

ex: { 4x + 3y = 10, 3x - y = 1 }

eq1 := 4*x + 3*y = 10;

eq2 := 3*x - y = 1;

solve ({eq1, eq2}, {x, y});

8) Define functions, compute derivatives, integrals, primitives.

f := x →

f := umapply (...., x);

} 2 correct ways of defining functions.

D → derivatives of functions

diff → derivatives of variables

- 1st order derivative of a function: $D(f)(x)$;
- 2nd order " " " " : $(D@@2)(f)(x)$;
- 1st order derivative of an expression (var): $\text{diff}(\text{expr}, x)$;
- 2nd order derivative of an expression (var): $\text{diff}(\text{expr}, x \$ 2)$;

$\boxed{\text{int}(f(x), x = -1 \dots 1)} \rightarrow \text{integral} \left(\int_{-1}^1 f(x) dx \right)$.

$\boxed{\text{int}(f(x), x)} \rightarrow \text{primitive}$.

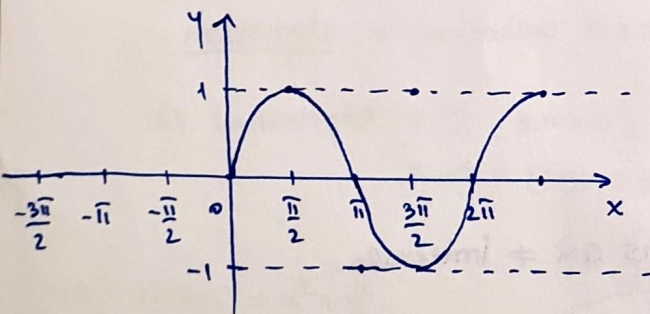
- derivative at a given point: $D(f)(\text{point})$;
- unnasign a function: $f(x) := 'f(x)'$.

9) ~ similar

10) Limits

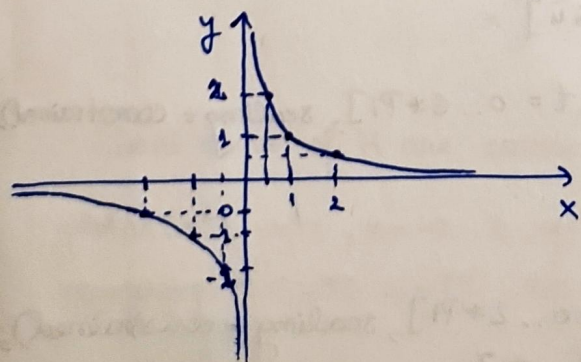
$\boxed{\text{limit}(\text{expr}, x = a)}$

11) $f(x) = \sin x$.



! open interval $(-8, 8)$: $\text{plot}(\sin(x), x = -8 + 10^1(-6) \dots 8 - 10^1(-6))$;

12) $f(x) = \frac{1}{x}$



$\text{plot}(1/x, x = -1 \dots 1, \text{discont} = \text{true})$;

13) Parametric curves

parametric plot syntax: $\text{plot}([x(t), y(t), t=a..b])$

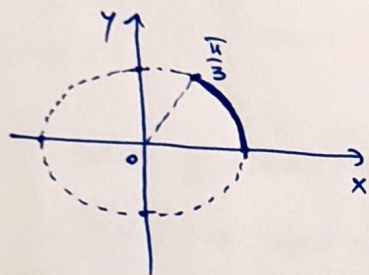
ex: $\text{plot}([2-t^2, t-t^3, t=-2..2, \text{scaling}=\text{constrained}]);$

↓
unit x = unit y (axis)

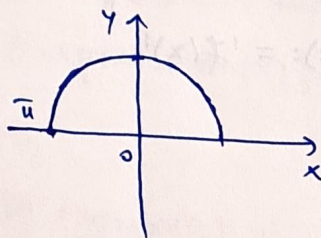
14) Parametric circle.

$$x = \cos(t), y = \sin(t)$$

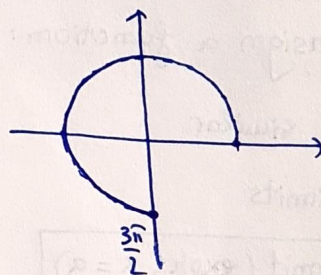
$$[0, \frac{\pi}{3}] \leftarrow t.$$



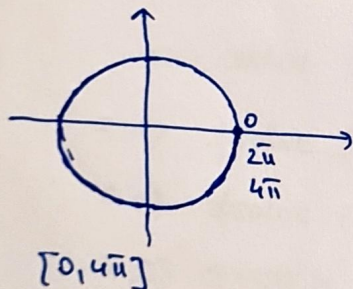
$$[0, \pi] \leftarrow t$$



$$[0, \frac{3\pi}{2}] \leftarrow t.$$



$$[0, 2\pi] \leftarrow t$$



→ same circle, but ≠ portion traced.

15) Planar curves of parametric equations on ≠ intervals.

a) $x = 2 \cos(\frac{t}{3})$; $y = 2 \cdot \sin(\frac{t}{3})$

⇒ circle radius = 2.

⇒ $\frac{t}{3} \Rightarrow$ slow rotation.

$$\frac{t}{3} \cdot 6\pi = 2\pi ; \quad \frac{t}{3} \in [0, 2\pi] ; \quad t \in [0, 6\pi]$$

⇒ $\text{plot}([2 * \cos(t/3), 2 * \sin(t/3), t = 0..6 * \pi], \text{scaling} = \text{constrained});$

b) $x = \cos(4t), y = \sin(4t)$

⇒ $4t \Rightarrow$ traces multiple times.

⇒ $\text{plot}([\cos(4 * t), \sin(4 * t), t = 0..2 * \pi], \text{scaling} = \text{constrained});$

⇒ traces the circle 4 times on $[0, 2\pi]$

1) Implicit equations of a circle centered at $(9, 10)$.

Plot ellipse, parabola, circle.

1) Circle w centre $(9, 10)$, radius r : $(x-9)^2 + (y-10)^2 = r^2$.

2) Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

3) Parabola: $y = x^2$.

1): centre $(9, 10)$, radius 3:

`implicitplot((x-9)^2 + (y-10)^2 = 9, x=4..14, y=5..15, sc=constr...);`

2) ellipse; $a=4$; $b=3$.

`implicitplot(x^2/16 + y^2/9 = 1, x=-6..6, y=-5..5, ...);`

3) parabola

`implicitplot(y = x^2, x=-3..3, y=0..10, ...);`

17) Planar curves of implicit equations.

implicit curve = all points satisfying equation.

rectangle = window $x=a..b$, $y=c..d$.

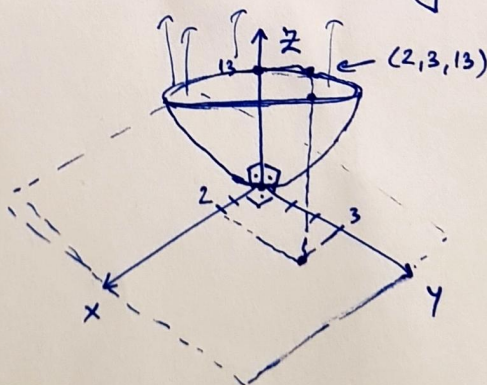
a) `implicitplot(x^2 - 2*x*y - y^2 = 1, x=-5..5, y=-5..5, grid=[250,250], scaling=constrained);`

18) $H(x, y) = x^2 + y^2$.

$$H(2, 3) = 13$$

$$H(0, 0) = 0$$

$$H(1, 1) = 2.$$



$z = x^2 + y^2$ (paraboloid)
 $x^2 + y^2 = c$ (circles centered at origin)

! level curves of H are concentric circles $x^2 + y^2 = c$

`plot3d(x^2 + y^2, x=-2..2, y=-2..2, axes=boxed);`
`contourplot(x^2 + y^2, x=-3..3, y=-3..3, contours=[1, 4, 9], ...);`

